PowerNile Graduate Project - SQL Phase

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- **Total Revenue by Sales Sector (Fuel vs Solar)**

Description: Calculates total revenue for fuel and solar sales sectors.

Code:

```
SELECT
   'Fuel Sales' AS sector,
   SUM(total_amount) AS total_revenue
FROM fuel_sales
UNION ALL
SELECT
   'Solar Sales' AS sector,
   SUM(total_price) AS total_revenue;
```

Key Findings:

1. Fuel stations serve as the primary revenue engine for the company, generating over 132.9M in overall sales.

2. The Solar sector delivers a robust performance with more than 44.1M in revenue, providing solid product diversification.

3. The combined revenue reflects a well-balanced revenue model split between core operational fuels and renewable energy solutions.

- **Top 3 Departments by Average Salary**

Description: Retrieves the top 3 departments with the highest average salary.

Code:

```
SELECT
   d.department_name,
   ROUND(AVG(e.salary), 2) AS avg_salary
FROM employees e
JOIN departments d ON e.department_id = d.department_id
GROUP BY d.department_name
ORDER BY avg_salary DESC
LIMIT 3;
```

Key Findings:

1. The highest average salaries are concentrated in specialized technical and operational departments.
2. Safety and Heavy Equipment departments lead the compensation tiers due to higher technical skill requirements and hazard exposure.
3. The compensation structure closely aligns with industry benchmarks for operational and field roles.

- **Monthly Sales Trend Comparison (Fuel vs Solar)**

Description: Compares monthly revenue trends between fuel and solar sectors.

Code:

```
SELECT
   c.month_name,
   c.month_num,
   COALESCE(SUM(f.total_amount), 0) AS fuel_revenue,
   COALESCE(SUM(s.total_price), 0) AS solar_revenue
FROM calendar c
LEFT JOIN fuel_sales f ON c.full_date = f.sale_date
LEFT JOIN solar_sales s ON c.full_date = s.sale_date
GROUP BY c.month_name, c.month_num
ORDER BY c.month_num;
```

Key Findings:

1. Fuel sales display consistent monthly cash flow with minor fluctuations during mid-year periods.

2. Solar sales experience demand surges during specific quarters, effectively offsetting fuel sector dips.
3. The monthly trend tracking enables precise cash-flow forecasting and inventory planning.

- **Real Estate Units Categorization**

Description: Categorizes real estate units based on unit price ranges.

Code:

```
SELECT
   unit_id,
   unit_type,
   unit_price,
   CASE
      WHEN unit_price < 200000 THEN 'Economy'
      WHEN unit_price <= 350000 THEN 'Mid-Tier'
      ELSE 'Luxury'
   END AS unit_category
FROM real_estate_units;
```

Key Findings:

1. Price segmentation categorizes property portfolios into clear market tiers (Economy, Mid-Tier, Luxury).
2. The classification assists marketing teams in targeting tailored buyer demographics.
3. Identifying price tiers highlights inventory distribution and potential absorption bottlenecks in higher-tier units.

- **High-Performing Employees (Sales > Department Average)**

Description: Identifies employees whose total sales exceed their department average.

Code:

```
WITH DeptAvgSales AS (

   SELECT
      e.department_id,
      AVG(f.total_amount) AS avg_dept_sales
   FROM fuel_sales f
   JOIN employees e ON f.employee_id = e.employee_id
   GROUP BY e.department_id
)
SELECT
   e.employee_id,
   e.first_name || ' ' || e.last_name AS employee_name,
   d.department_name,
   SUM(f.total_amount) AS total_employee_sales
FROM fuel_sales f
JOIN employees e ON f.employee_id = e.employee_id
JOIN departments d ON e.department_id = d.department_id
JOIN DeptAvgSales das ON e.department_id = das.department_id
GROUP BY e.employee_id, e.first_name, e.last_name, d.department_name,
das.avg_dept_sales
HAVING SUM(f.total_amount) > das.avg_dept_sales;
```

Key Findings:

1. This query isolates top sales talent whose individual throughput surpasses department benchmarks.
2. Identifying outperformers allows executive management to structure performance incentives and retention programs.
3. Sales distribution patterns highlight key contributors driving overall divisional performance.

- **Total Real Estate Revenue by Category**

Description: Groups and sums total revenue and unit counts by real estate category.

Code:

```
SELECT
   CASE
      WHEN unit_price < 200000 THEN 'Economy'
      WHEN unit_price <= 350000 THEN 'Mid-Tier'
      ELSE 'Luxury'
   END AS unit_category,
   COUNT(unit_id) AS total_units,
   SUM(unit_price) AS total_revenue
FROM real_estate_units
GROUP BY unit_category
ORDER BY total_revenue DESC;
```

Key Findings:

1. Highlights which property tier delivers the largest financial contribution to total real estate revenue.
2. Compares unit inventory volume against aggregate financial value across categories.
3. Provides data-driven insights for strategic real estate development and pricing adjustments.

- **Employee Count and Salary Distribution per Department**

Description: Evaluates employee headcount, total salary expense, and average salary per department.

Code:

```
SELECT
   d.department_name,
   COUNT(e.employee_id) AS total_employees,
   SUM(e.salary) AS total_salary_expense,
   ROUND(AVG(e.salary), 2) AS average_salary
FROM departments d
LEFT JOIN employees e ON d.department_id = e.department_id
GROUP BY d.department_name
ORDER BY total_employees DESC;
```

Key Findings:
1. Operational divisions such as Mining and Energy contain the highest workforce concentration.

2. Outlines payroll expenses per department to ensure budget alignment and labor efficiency.
3. Serves as a baseline for HR organizational planning and resource allocation.

- **Top Customer Sales in Solar Energy**

Description: Identifies the top 5 customers with highest total spending in solar energy.

Code:

```
SELECT
   s.customer_name,
   COUNT(s.sale_id) AS total_orders,
   SUM(s.total_price) AS total_spent
FROM solar_sales s
GROUP BY s.customer_name
ORDER BY total_spent DESC
LIMIT 5;
```

Key Findings:
1. Demonstrates a high reliance on key institutional and corporate clients in the solar division.

2. The top 5 customers account for a significant share of total solar energy equipment revenue.

3. Recommends implementing Key Account Management (KAM) programs to maintain long-term client retention.